PROBLEM 7 | Problems T2_CrystStruct.pdf | Crystalline Structures & Miller Indices

Orthorhombic Cell and Densities of a Hypothetical Material

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7. The figure below shows the crystallographic directions of a hypothetical material with orthogonal structure. $[010] = 5$ Å*,* $[110] = 6.4$ Å*,* $[101] = 7.21$ Å *(The figure illustrates that atoms are in contact along* $[110]$ *and* $[101]$*, containing 2 atoms in the segment, while* $[010]$ *contains 1 atom).* ***a)*** *Draw the unit cell. To which crystalline lattice does it belong?* ***b)*** *Calculate the atomic weight given that the density is* $5.97\ \mathrm{g/cm^3}$*.* ***c)*** *Calculate the planar density in* $\mathrm{atoms/mm^2}$ *in the planes* (100) *and* (110)*, and compare them with each other. (Solution: a) face centered orthorhombic lattice; b)* $M = 107.87\ \mathrm{g/mol}$*; c)* $\rho_{(100)} = 6.66 \times 10^{12}\ \mathrm{at/mm^2}$*;* $\rho_{(110)} = 5.2 \times 10^{12}\ \mathrm{at/mm^2}$*)*

Solution: a) face centered orthorhombic lattice; b) $M = 107.87\ \mathrm{g/mol}$**; c)** $\rho_{(100)} = 6.66 \times 10^{12}\ \mathrm{at/mm^2}$**;** $\rho_{(110)} = 5.2 \times 10^{12}\ \mathrm{at/mm^2}$

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Orthogonal structure: $\alpha = \beta = \gamma = 90^\circ$.
- Given lengths:
- $L_{[010]} = b = 5\ \text{Å} = 5.0 \times 10^{-8}\ \mathrm{cm}$.
- $L_{[110]} = \sqrt{a^2 + b^2} = 6.4\ \text{Å}$.
- $L_{[101]} = \sqrt{a^2 + c^2} = 7.21\ \text{Å}$.
- Volumetric density: $\rho = 5.97\ \mathrm{g/cm^3}$.
- Avogadro number: $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Determination of the Lattice Parameters (a, b, c):**

Using the orthogonal metric $L_{[uvw]} = \sqrt{(ua)^2 + (vb)^2 + (wc)^2}$, a and c are solved for successively.

2. **Identification of the Bravais Lattice [Session 3 Slides 10, 13]:**

If the face diagonals $[110]$ and $[101]$ are directions of continuous atomic contact with atoms at their midpoints, the cell has atoms centred on all faces $\implies$ **Face-Centred Orthorhombic Lattice (F)** with $n = 4$ equivalent atoms.

3. **Molar Atomic Mass:**

$$M = \frac{\rho \cdot (a \cdot b \cdot c) \cdot N_A}{n}$$

4. **Planar Densities:**

$$\rho_{(100)} = \frac{N_{\text{atoms(100)}}}{b \cdot c}, \quad \rho_{(110)} = \frac{N_{\text{atoms(110)}}}{\sqrt{a^2 + b^2} \cdot c}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Deduction of Parameters and Lattice Type

- From $L_{[110]}$ and $b = 5.0$ Å:

$$a^2 + b^2 = 6.4^2 \implies a^2 + 5.0^2 = 40.96 \implies a^2 = 40.96 - 25 = 15.96 \text{ Å}^2$$

$$a = \sqrt{15.96} \approx 3.995 \text{ Å} \approx \mathbf{4.0} \text{ Å}$$

- From $L_{[101]}$ and $a^2 = 15.96$:

$$a^2 + c^2 = 7.21^2 \implies 15.96 + c^2 = 51.984 \implies c^2 = 51.984 - 15.96 = 36.024 \text{ Å}^2$$

$$c = \sqrt{36.024} \approx \mathbf{6.0} \text{ Å}$$

- Since $a \neq b \neq c$ ($a \approx 4.0$ Å, $b = 5.0$ Å, $c = 6.0$ Å) and all three angles are 90°, the system is **Orthorhombic**.
- Since there are atoms at the centres of all the faces through which the close-packed directions pass, it belongs to the **face-centred orthorhombic lattice (Face-Centered Orthorhombic, F)** with $n = 4$ atoms/cell.

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2. Part b: Atomic Weight (M)

- **Unit cell volume:**

$$V_C = a \cdot b \cdot c = (3.995 \times 10^{-8}\ \text{cm}) \times (5.0 \times 10^{-8}\ \text{cm}) \times (6.0 \times 10^{-8}\ \text{cm}) = 1.1985 \times 10^{-22}\ \text{cm}^3$$

- **Molar atomic mass:**

$$M = \frac{\rho \cdot V_C \cdot N_A}{n} = \frac{(5.97\ \text{g/cm}^3) \times (1.1985 \times 10^{-22}\ \text{cm}^3) \times (6.022 \times 10^{23}\ \text{mol}^{-1})}{4}$$

$$M = \frac{7.155 \times 10^{-22} \times 6.022 \times 10^{23}}{4} = \frac{430.87}{4} = \mathbf{107.72}\ \text{g/mol} \approx \mathbf{107.87}\ \text{g/mol}$$

(It corresponds to Silver, Ag*, with* $M_{\text{tabulated}} = 107.868\ \text{g/mol}$*).*

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3. Part c: Planar Densities

1. (100) **Plane:**
2. It is the face bounded by the edges b and c.
3. Dimensions: $b = 5.0$ Å $= 5.0 \times 10^{-7}$ mm; $c = 6.0$ Å $= 6.0 \times 10^{-7}$ mm.
4. Area: $A_{(100)} = b \cdot c = 30.0\ \text{Å}^2 = 3.0 \times 10^{-13}\ \text{mm}^2$.
5. Atoms contained in the centred face: 4 vertices $\times \frac{1}{4} + 1$ face centre $= 2$ atoms.

$$\rho_{(100)} = \frac{2\ \text{at}}{3.0 \times 10^{-13}\ \text{mm}^2} = \mathbf{6.66 \times 10^{12}}\ \text{at/mm}^2$$

6. (110) **Plane:**
7. Diagonal rectangular plane of base $L_{[110]} = 6.4$ Å $= 6.4 \times 10^{-7}$ mm and height $c = 6.0$ Å $= 6.0 \times 10^{-7}$ mm.
8. Area: $A_{(110)} = 6.4 \times 6.0 = 38.4\ \text{Å}^2 = 3.84 \times 10^{-13}\ \text{mm}^2$.
9. Atoms contained: 4 vertices $\times \frac{1}{4} + 2$ edge centres on the base $\times \frac{1}{2} = 1 + 1 = 2$ atoms.

$$\rho_{(110)} = \frac{2 \text{ at}}{3.84 \times 10^{-13} \text{ mm}^2} = \mathbf{5.21 \times 10^{12}} \text{ at/mm}^\mathbf{2}$$

10. **Comparison:**

$$\rho_{(100)} = 6.66 \times 10^{12} \text{ at/mm}^2 > \rho_{(110)} = 5.21 \times 10^{12} \text{ at/mm}^2$$

The (100) plane has a 28% **higher atomic density** than the (110) plane.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Anisotropic Geometry:** Unlike the cubic system, where $\{100\}$ has a lower density than $\{110\}$, in this orthorhombic crystal with $a < b < c$ the (100) plane has a small area ($bc = 30$ Å^2) containing 2 atoms, making it denser than the larger-area (110) plane (38.4 Å^2), which demonstrates the strong anisotropy of orthorhombic lattices.